Table 1: Detailed instance-level results for Next Release Problem, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
nrp-e1	1.0	2.0	3.0	1.0	0.002 ± 0.002	0.908 ± 0.040	0.962 ± 0.020	0.920 ± 0.017
nrp-e2	1.0	2.0	3.0	1.0	0.001 ± 0.001	0.931 ± 0.009	0.977 ± 0.017	0.929 ± 0.024
nrp-e3	1.0	2.0	3.0	1.0	0.000 ± 0.000	0.898 ± 0.021	0.958 ± 0.039	0.901 ± 0.033
nrp-e4	1.0	2.0	4.0	3.0	0.000 ± 0.001	0.902 ± 0.020	0.967 ± 0.021	0.931 ± 0.025
nrp-g1	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.907 ± 0.020	0.971 ± 0.015	0.915 ± 0.029
nrp-g2	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.905 ± 0.017	0.948 ± 0.044	0.913 ± 0.025
nrp-g3	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.902 ± 0.021	0.957 ± 0.041	0.924 ± 0.034
nrp-g4	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.868 ± 0.065	0.934 ± 0.054	0.898 ± 0.041
nrp-m1	1.0	2.0	4.0	3.0	0.001 ± 0.002	0.924 ± 0.013	0.969 ± 0.013	0.929 ± 0.019
nrp-m2	1.0	2.0	3.0	1.0	0.001 ± 0.001	0.928 ± 0.021	0.974 ± 0.019	0.930 ± 0.027
nrp-m3	1.0	3.0	4.0	2.0	0.001 ± 0.001	0.955 ± 0.014	0.981 ± 0.013	0.941 ± 0.021
nrp-m4	1.0	3.0	4.0	2.0	0.001 ± 0.001	0.931 ± 0.019	0.969 ± 0.022	0.928 ± 0.016
nrp1	1.0	3.0	4.0	2.0	0.000 ± 0.000	0.894 ± 0.048	0.916 ± 0.081	0.871 ± 0.041
nrp2	1.0	4.0	3.0	2.0	0.027 ± 0.017	0.976 ± 0.017	0.903 ± 0.035	0.865 ± 0.023
nrp3	1.0	2.0	4.0	3.0	0.003 ± 0.002	0.937 ± 0.012	0.981 ± 0.011	0.947 ± 0.016
nrp4	1.0	2.0	3.0	1.0	0.000 ± 0.000	0.955 ± 0.011	0.983 ± 0.013	0.957 ± 0.009
nrp5	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.950 ± 0.012	0.982 ± 0.020	0.958 ± 0.010
Average	**1.00**	**2.29**	**3.65**	**2.18**	**0.002 ± 0.002**	**0.922 ± 0.022**	**0.961 ± 0.028**	**0.921 ± 0.024**

Table 2: Detailed instance-level results for Software Configuration Tuning, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
dnn_adiac	1.0	4.0	2.0	3.0	0.343 ± 0.370	0.781 ± 0.244	0.628 ± 0.265	0.657 ± 0.302
dnn_adiac_reverse	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.075 ± 0.061	0.181 ± 0.291	0.103 ± 0.044
dnn_coffee	1.0	1.0	3.0	2.0	0.000 ± 0.000	0.386 ± 0.415	0.300 ± 0.389	0.286 ± 0.369
dnn_coffee_reverse	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.135 ± 0.108	0.544 ± 0.388	0.457 ± 0.280
dnn_dsr	1.0	2.0	1.0	1.0	0.004 ± 0.012	0.162 ± 0.306	0.205 ± 0.249	0.236 ± 0.376
dnn_dsr_reverse	1.0	2.0	3.0	3.0	0.000 ± 0.000	0.089 ± 0.089	0.328 ± 0.289	0.319 ± 0.214
dnn_sa	1.0	3.0	2.0	1.0	0.028 ± 0.088	0.779 ± 0.290	0.699 ± 0.220	0.731 ± 0.230
dnn_sa_reverse	1.0	2.0	3.0	3.0	0.000 ± 0.000	0.480 ± 0.120	0.623 ± 0.285	0.623 ± 0.199
llvm	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.322 ± 0.271	0.830 ± 0.133	0.687 ± 0.230
llvm_reverse	1.0	2.0	3.0	1.0	0.134 ± 0.072	0.551 ± 0.214	0.678 ± 0.199	0.512 ± 0.259
lrzip	1.0	3.0	2.0	4.0	0.220 ± 0.167	0.453 ± 0.281	0.317 ± 0.222	0.579 ± 0.272
lrzip_reverse	1.0	4.0	2.0	3.0	0.000 ± 0.000	0.328 ± 0.306	0.037 ± 0.042	0.094 ± 0.080
mariadb	2.0	2.0	1.0	3.0	0.000 ± 0.000	0.121 ± 0.254	0.000 ± 0.000	0.430 ± 0.386
mariadb_reverse	1.0	3.0	2.0	2.0	0.000 ± 0.000	0.332 ± 0.297	0.289 ± 0.348	0.167 ± 0.219
mongodb	1.0	1.0	2.0	3.0	0.016 ± 0.049	0.071 ± 0.158	0.331 ± 0.319	0.528 ± 0.307
mongodb_reverse	1.0	1.0	2.0	3.0	0.011 ± 0.035	0.022 ± 0.047	0.449 ± 0.291	0.568 ± 0.298
storm_rs	1.0	2.0	3.0	3.0	0.000 ± 0.000	0.442 ± 0.312	0.567 ± 0.394	0.658 ± 0.355
storm_rs_reverse	1.0	1.0	2.0	3.0	0.000 ± 0.000	0.015 ± 0.046	0.259 ± 0.316	0.378 ± 0.299
storm_wc	1.0	3.0	2.0	4.0	0.000 ± 0.000	0.248 ± 0.327	0.173 ± 0.110	0.260 ± 0.163
storm_wc_reverse	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.101 ± 0.136	0.392 ± 0.242	0.284 ± 0.082
trimesh	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
trimesh_reverse	1.0	3.0	2.0	2.0	0.036 ± 0.030	0.708 ± 0.175	0.669 ± 0.225	0.617 ± 0.270
vp9	1.0	2.0	3.0	3.0	0.000 ± 0.000	0.163 ± 0.036	0.198 ± 0.312	0.166 ± 0.150
vp9_reverse	1.0	2.0	1.0	1.0	0.058 ± 0.121	0.410 ± 0.364	0.326 ± 0.391	0.329 ± 0.357
x264	2.0	2.0	1.0	3.0	0.000 ± 0.000	0.887 ± 0.240	0.000 ± 0.000	0.888 ± 0.022
x264_reverse	2.0	2.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.590 ± 0.162
Average	**1.12**	**2.15**	**2.31**	**2.54**	**0.052 ± 0.036**	**0.329 ± 0.196**	**0.366 ± 0.228**	**0.448 ± 0.228**

Table 3: Detailed instance-level results for Software Effort Estimation, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
china_fold1	1.0	2.0	3.0	1.0	0.077 ± 0.052	0.640 ± 0.120	0.709 ± 0.193	0.643 ± 0.157
china_fold2	1.0	3.0	4.0	2.0	0.020 ± 0.023	0.784 ± 0.140	0.830 ± 0.100	0.721 ± 0.092
china_fold3	1.0	2.0	3.0	1.0	0.004 ± 0.004	0.362 ± 0.198	0.425 ± 0.240	0.345 ± 0.179
desharnais_fold1	1.0	3.0	2.0	1.0	0.128 ± 0.120	0.792 ± 0.151	0.651 ± 0.200	0.643 ± 0.114
desharnais_fold2	1.0	2.0	3.0	3.0	0.068 ± 0.052	0.607 ± 0.196	0.641 ± 0.151	0.655 ± 0.228
desharnais_fold3	1.0	3.0	2.0	1.0	0.098 ± 0.074	0.661 ± 0.181	0.618 ± 0.129	0.667 ± 0.169
finnish_fold1	1.0	4.0	2.0	3.0	0.031 ± 0.024	0.564 ± 0.274	0.321 ± 0.163	0.349 ± 0.105
finnish_fold2	1.0	3.0	2.0	1.0	0.057 ± 0.067	0.624 ± 0.188	0.540 ± 0.179	0.514 ± 0.129
finnish_fold3	1.0	2.0	1.0	1.0	0.017 ± 0.013	0.720 ± 0.164	0.732 ± 0.110	0.766 ± 0.182
maxwell_fold1	1.0	3.0	2.0	2.0	0.027 ± 0.022	0.875 ± 0.066	0.861 ± 0.077	0.801 ± 0.061
maxwell_fold2	1.0	3.0	2.0	2.0	0.043 ± 0.028	0.893 ± 0.066	0.894 ± 0.065	0.821 ± 0.073
maxwell_fold3	1.0	3.0	2.0	2.0	0.048 ± 0.040	0.867 ± 0.083	0.892 ± 0.071	0.790 ± 0.030
miyazaki_fold1	1.0	3.0	2.0	2.0	0.004 ± 0.002	0.637 ± 0.207	0.478 ± 0.150	0.622 ± 0.207
miyazaki_fold2	1.0	3.0	2.0	1.0	0.003 ± 0.002	0.530 ± 0.306	0.419 ± 0.125	0.574 ± 0.215
miyazaki_fold3	1.0	4.0	2.0	3.0	0.003 ± 0.001	0.602 ± 0.326	0.387 ± 0.146	0.520 ± 0.181
Average	**1.00**	**2.87**	**2.27**	**1.73**	**0.042 ± 0.035**	**0.677 ± 0.178**	**0.627 ± 0.140**	**0.629 ± 0.142**

Table 4: Detailed instance-level results for Software Defect Prediction, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
ant-1.7_J48	1.0	3.0	2.0	4.0	0.101 ± 0.067	0.543 ± 0.121	0.456 ± 0.170	0.617 ± 0.213
ant-1.7_KNN	1.0	2.0	1.0	1.0	0.133 ± 0.067	0.473 ± 0.156	0.503 ± 0.169	0.471 ± 0.241
ant-1.7_LR	1.0	2.0	1.0	1.0	0.380 ± 0.128	0.585 ± 0.298	0.588 ± 0.272	0.569 ± 0.290
ant-1.7_NB	1.0	2.0	1.0	1.0	0.200 ± 0.108	0.511 ± 0.258	0.516 ± 0.298	0.473 ± 0.264
camel-1.6_J48	1.0	3.0	2.0	2.0	0.094 ± 0.071	0.662 ± 0.199	0.642 ± 0.217	0.578 ± 0.276
camel-1.6_KNN	1.0	2.0	3.0	1.0	0.133 ± 0.067	0.635 ± 0.179	0.685 ± 0.253	0.573 ± 0.257
camel-1.6_LR	1.0	2.0	1.0	1.0	0.175 ± 0.101	0.450 ± 0.294	0.470 ± 0.295	0.458 ± 0.334
camel-1.6_NB	1.0	2.0	1.0	1.0	0.206 ± 0.052	0.322 ± 0.164	0.279 ± 0.167	0.336 ± 0.262
ivy-2.0_J48	1.0	2.0	1.0	1.0	0.282 ± 0.138	0.750 ± 0.115	0.785 ± 0.137	0.751 ± 0.141
ivy-2.0_KNN	1.0	3.0	2.0	2.0	0.166 ± 0.087	0.561 ± 0.173	0.565 ± 0.150	0.528 ± 0.227
ivy-2.0_LR	1.0	2.0	1.0	1.0	0.110 ± 0.049	0.342 ± 0.170	0.344 ± 0.254	0.359 ± 0.195
ivy-2.0_NB	1.0	2.0	3.0	1.0	0.172 ± 0.075	0.487 ± 0.236	0.540 ± 0.292	0.549 ± 0.205
jedit-4.0_J48	1.0	2.0	3.0	3.0	0.112 ± 0.069	0.575 ± 0.235	0.683 ± 0.222	0.628 ± 0.179
jedit-4.0_KNN	1.0	2.0	1.0	1.0	0.200 ± 0.087	0.686 ± 0.181	0.650 ± 0.176	0.660 ± 0.179
jedit-4.0_LR	1.0	2.0	1.0	1.0	0.140 ± 0.099	0.613 ± 0.252	0.630 ± 0.182	0.641 ± 0.211
jedit-4.0_NB	1.0	2.0	3.0	3.0	0.135 ± 0.070	0.508 ± 0.121	0.557 ± 0.251	0.542 ± 0.183
lucene-2.4_J48	1.0	3.0	2.0	4.0	0.127 ± 0.070	0.584 ± 0.181	0.486 ± 0.171	0.593 ± 0.116
lucene-2.4_KNN	1.0	2.0	1.0	1.0	0.152 ± 0.083	0.484 ± 0.261	0.436 ± 0.218	0.465 ± 0.146
lucene-2.4_LR	1.0	2.0	1.0	1.0	0.373 ± 0.086	0.560 ± 0.258	0.527 ± 0.208	0.554 ± 0.249
lucene-2.4_NB	1.0	2.0	1.0	1.0	0.436 ± 0.091	0.624 ± 0.245	0.627 ± 0.286	0.590 ± 0.317
poi-3.0_J48	1.0	3.0	2.0	2.0	0.122 ± 0.080	0.578 ± 0.156	0.557 ± 0.192	0.516 ± 0.273
poi-3.0_KNN	1.0	2.0	3.0	3.0	0.202 ± 0.091	0.564 ± 0.238	0.575 ± 0.192	0.570 ± 0.194
poi-3.0_LR	1.0	2.0	3.0	1.0	0.264 ± 0.112	0.450 ± 0.228	0.458 ± 0.231	0.529 ± 0.159
poi-3.0_NB	1.0	2.0	3.0	1.0	0.242 ± 0.111	0.487 ± 0.213	0.513 ± 0.207	0.471 ± 0.185
synapse-1.2_J48	1.0	2.0	3.0	4.0	0.102 ± 0.083	0.471 ± 0.193	0.531 ± 0.191	0.661 ± 0.202
synapse-1.2_KNN	1.0	2.0	1.0	1.0	0.130 ± 0.094	0.589 ± 0.220	0.513 ± 0.256	0.562 ± 0.205
synapse-1.2_LR	1.0	2.0	1.0	1.0	0.216 ± 0.118	0.583 ± 0.248	0.607 ± 0.217	0.605 ± 0.206
synapse-1.2_NB	1.0	2.0	1.0	1.0	0.197 ± 0.118	0.574 ± 0.299	0.578 ± 0.228	0.599 ± 0.245
velocity-1.6_J48	1.0	3.0	2.0	1.0	0.066 ± 0.042	0.547 ± 0.168	0.485 ± 0.216	0.446 ± 0.111
velocity-1.6_KNN	1.0	3.0	2.0	1.0	0.163 ± 0.103	0.665 ± 0.078	0.523 ± 0.227	0.553 ± 0.256
velocity-1.6_LR	1.0	2.0	3.0	1.0	0.202 ± 0.076	0.347 ± 0.259	0.371 ± 0.264	0.410 ± 0.230
velocity-1.6_NB	1.0	1.0	2.0	2.0	0.306 ± 0.077	0.420 ± 0.217	0.471 ± 0.266	0.314 ± 0.186
xalan-2.6_J48	1.0	4.0	2.0	3.0	0.173 ± 0.098	0.640 ± 0.345	0.538 ± 0.240	0.613 ± 0.231
xalan-2.6_KNN	1.0	2.0	3.0	1.0	0.134 ± 0.120	0.579 ± 0.206	0.675 ± 0.245	0.512 ± 0.288
xalan-2.6_LR	1.0	2.0	1.0	1.0	0.339 ± 0.134	0.414 ± 0.300	0.419 ± 0.305	0.411 ± 0.307
xalan-2.6_NB	1.0	2.0	3.0	4.0	0.375 ± 0.106	0.447 ± 0.253	0.475 ± 0.206	0.544 ± 0.237
xerces-1.4_J48	2.0	1.0	2.0	2.0	0.249 ± 0.110	0.293 ± 0.229	0.354 ± 0.314	0.378 ± 0.288
xerces-1.4_KNN	1.0	2.0	3.0	1.0	0.131 ± 0.123	0.393 ± 0.183	0.490 ± 0.259	0.409 ± 0.176
xerces-1.4_LR	1.0	1.0	2.0	2.0	0.380 ± 0.097	0.415 ± 0.233	0.478 ± 0.272	0.456 ± 0.224
xerces-1.4_NB	1.0	2.0	3.0	1.0	0.293 ± 0.129	0.431 ± 0.292	0.526 ± 0.322	0.440 ± 0.311
Average	**1.02**	**2.15**	**1.93**	**1.65**	**0.203 ± 0.092**	**0.521 ± 0.216**	**0.528 ± 0.231**	**0.523 ± 0.225**

Table 5: Detailed instance-level results for Software Product Lines Testing, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	Rank				Mean ± Std			
	$\mathbf{MMO}_b$	SWAY	LINE	LITE	$\mathbf{MMO}_b$	SWAY	LINE	LITE
7z	1.0	4.0	2.0	3.0	0.005 ± 0.004	0.691 ± 0.174	0.347 ± 0.075	0.525 ± 0.137
amazon	1.0	4.0	2.0	3.0	0.014 ± 0.007	0.818 ± 0.126	0.528 ± 0.065	0.634 ± 0.099
berkeleydbc	1.0	4.0	2.0	3.0	0.019 ± 0.020	0.778 ± 0.131	0.441 ± 0.090	0.593 ± 0.049
cocheecologico	1.0	4.0	2.0	3.0	0.018 ± 0.011	0.760 ± 0.127	0.522 ± 0.059	0.619 ± 0.108
counterstrikesimplefeatu	1.0	3.0	2.0	1.0	0.036 ± 0.020	0.802 ± 0.145	0.571 ± 0.070	0.602 ± 0.093
drupal	1.0	4.0	2.0	3.0	0.031 ± 0.026	0.871 ± 0.099	0.648 ± 0.054	0.764 ± 0.080
dssample	1.0	4.0	2.0	3.0	0.014 ± 0.014	0.830 ± 0.094	0.256 ± 0.039	0.539 ± 0.156
dune	1.0	4.0	2.0	3.0	0.041 ± 0.027	0.886 ± 0.067	0.548 ± 0.070	0.717 ± 0.084
electronicdrum	1.0	4.0	2.0	3.0	0.049 ± 0.035	0.814 ± 0.113	0.536 ± 0.068	0.740 ± 0.076
hipacc	1.0	4.0	2.0	3.0	0.026 ± 0.022	0.839 ± 0.122	0.592 ± 0.042	0.667 ± 0.068
javagc	1.0	4.0	2.0	3.0	0.050 ± 0.024	0.853 ± 0.126	0.562 ± 0.077	0.745 ± 0.135
jhipster	1.0	4.0	2.0	3.0	0.029 ± 0.019	0.901 ± 0.097	0.531 ± 0.074	0.640 ± 0.128
lrzip	2.0	4.0	1.0	3.0	0.000 ± 0.000	0.615 ± 0.252	0.000 ± 0.000	0.045 ± 0.093
modeltransformation	1.0	3.0	2.0	2.0	0.043 ± 0.031	0.852 ± 0.113	0.660 ± 0.065	0.866 ± 0.061
polly	1.0	4.0	2.0	3.0	0.035 ± 0.018	0.859 ± 0.115	0.466 ± 0.066	0.778 ± 0.185
smarthomev2.2	1.0	4.0	2.0	3.0	0.053 ± 0.031	0.852 ± 0.117	0.580 ± 0.061	0.667 ± 0.073
splssimuelespnp	1.0	4.0	2.0	3.0	0.021 ± 0.013	0.755 ± 0.174	0.465 ± 0.064	0.626 ± 0.089
videoplayer	1.0	4.0	2.0	3.0	0.037 ± 0.028	0.869 ± 0.094	0.653 ± 0.070	0.746 ± 0.087
vp9	1.0	4.0	2.0	3.0	0.027 ± 0.013	0.785 ± 0.207	0.472 ± 0.054	0.655 ± 0.132
webportal	1.0	4.0	2.0	3.0	0.055 ± 0.040	0.879 ± 0.089	0.661 ± 0.054	0.766 ± 0.083
x264	1.0	4.0	2.0	3.0	0.065 ± 0.034	0.790 ± 0.151	0.465 ± 0.075	0.570 ± 0.092
Average	**1.05**	**3.90**	**1.95**	**2.86**	$\mathbf{0.032 \pm 0.021}$	$\mathbf{0.814 \pm 0.130}$	$\mathbf{0.500 \pm 0.061}$	$\mathbf{0.643 \pm 0.100}$

Table 6: Detailed instance-level results for Software Project Scheduling, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better). × indicates that the algorithm failed to obtain valid solutions on that instance.

Instance	Rank				Mean ± Std			
	$\mathbf{MMO}_b$	SWAY	LINE	LITE	$\mathbf{MMO}_b$	SWAY	LINE	LITE
10-10-skill-4-5	1.0	2.0	3.0	1.0	0.002 ± 0.002	0.884 ± 0.025	0.963 ± 0.022	0.893 ± 0.021
10-10-skill-6-7	1.0	2.0	4.0	3.0	0.004 ± 0.002	0.898 ± 0.024	0.977 ± 0.015	0.916 ± 0.012
10-15-skill-4-5	1.0	2.0	3.0	1.0	0.001 ± 0.001	0.929 ± 0.008	0.960 ± 0.025	0.927 ± 0.008
10-15-skill-6-7	1.0	2.0	4.0	3.0	0.000 ± 0.000	0.956 ± 0.008	0.984 ± 0.013	0.962 ± 0.007
10-5-skill-4-5	1.0	2.0	3.0	1.0	0.000 ± 0.000	0.923 ± 0.016	0.981 ± 0.016	0.919 ± 0.024
10-5-skill-6-7	1.0	2.0	4.0	3.0	0.001 ± 0.001	0.878 ± 0.039	0.951 ± 0.040	0.901 ± 0.016
20-10-skill-4-5	1.0	3.0	4.0	2.0	0.002 ± 0.001	0.940 ± 0.017	0.982 ± 0.017	0.935 ± 0.014
20-10-skill-6-7	1.0	2.0	4.0	3.0	0.001 ± 0.001	0.940 ± 0.014	0.982 ± 0.011	0.946 ± 0.015
20-15-skill-4-5	1.0	2.0	3.0	1.0	0.003 ± 0.002	0.937 ± 0.006	0.971 ± 0.020	0.940 ± 0.010
20-15-skill-6-7	1.0	3.0	4.0	2.0	0.005 ± 0.005	0.920 ± 0.009	0.963 ± 0.033	0.910 ± 0.013
20-5-skill-4-5	1.0	3.0	4.0	2.0	0.003 ± 0.002	0.873 ± 0.028	0.950 ± 0.032	0.868 ± 0.025
20-5-skill-6-7	1.0	2.0	3.0	1.0	0.002 ± 0.001	0.895 ± 0.025	0.956 ± 0.031	0.897 ± 0.026
30-10-skill-4-5	1.0	2.0	4.0	3.0	0.001 ± 0.001	0.907 ± 0.014	0.955 ± 0.032	0.915 ± 0.016
30-10-skill-6-7	1.0	2.0	4.0	3.0	0.003 ± 0.002	0.929 ± 0.013	0.975 ± 0.017	0.936 ± 0.015
30-15-skill-4-5	1.0	2.0	4.0	3.0	0.002 ± 0.001	0.896 ± 0.014	0.969 ± 0.026	0.898 ± 0.021
30-15-skill-6-7	1.0	2.0	3.0	1.0	0.005 ± 0.003	0.921 ± 0.019	0.968 ± 0.021	0.920 ± 0.017
30-5-skill-4-5	1.0	2.0	3.0	1.0	0.004 ± 0.002	0.909 ± 0.042	×	0.912 ± 0.043
30-5-skill-6-7	1.0	3.0	4.0	2.0	0.001 ± 0.001	0.878 ± 0.024	0.962 ± 0.023	0.860 ± 0.016
Average	**1.00**	**2.22**	**3.61**	**2.00**	$\mathbf{0.002 \pm 0.002}$	$\mathbf{0.912 \pm 0.019}$	$\mathbf{0.969 \pm 0.022}$	$\mathbf{0.914 \pm 0.018}$

Table 7: Detailed instance-level results for Third Party Library Migration, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
migrationRule10_ALL	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule10_CO	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule10_DS	1.0	2.0	3.0	3.0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.200 ± 0.422
migrationRule10_MS	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_ALL	1.0	4.0	2.0	3.0	0.000 ± 0.000	0.618 ± 0.294	0.272 ± 0.185	0.459 ± 0.221
migrationRule18_CO	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_DS	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_MS	1.0	3.0	2.0	2.0	0.000 ± 0.000	0.414 ± 0.418	0.143 ± 0.213	0.372 ± 0.364
migrationRule1_ALL	1.0	4.0	3.0	2.0	0.017 ± 0.016	0.929 ± 0.080	0.831 ± 0.080	0.785 ± 0.123
migrationRule1_CO	1.0	4.0	3.0	2.0	0.008 ± 0.011	0.891 ± 0.109	0.858 ± 0.125	0.781 ± 0.141
migrationRule1_DS	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule1_MS	1.0	2.0	1.0	1.0	0.000 ± 0.000	0.740 ± 0.171	0.740 ± 0.117	0.730 ± 0.134
migrationRule2_ALL	1.0	3.0	2.0	2.0	0.041 ± 0.041	0.929 ± 0.052	0.939 ± 0.039	0.913 ± 0.034
migrationRule2_CO	1.0	3.0	2.0	2.0	0.072 ± 0.071	0.938 ± 0.063	0.947 ± 0.047	0.912 ± 0.055
migrationRule2_DS	1.0	4.0	3.0	2.0	0.056 ± 0.035	0.960 ± 0.038	0.929 ± 0.040	0.894 ± 0.039
migrationRule2_MS	1.0	4.0	3.0	2.0	0.082 ± 0.056	0.921 ± 0.053	0.875 ± 0.078	0.836 ± 0.100
migrationRule3_ALL	1.0	3.0	2.0	1.0	0.109 ± 0.167	0.831 ± 0.107	0.711 ± 0.322	0.707 ± 0.127
migrationRule3_CO	1.0	3.0	4.0	2.0	0.319 ± 0.280	0.798 ± 0.228	0.818 ± 0.241	0.600 ± 0.238
migrationRule3_MS	1.0	3.0	4.0	2.0	0.033 ± 0.105	0.600 ± 0.141	0.667 ± 0.157	0.500 ± 0.176
migrationRule4_ALL	1.0	3.0	2.0	1.0	0.007 ± 0.021	0.845 ± 0.130	0.763 ± 0.066	0.754 ± 0.086
migrationRule4_CO	1.0	4.0	3.0	2.0	0.000 ± 0.000	0.767 ± 0.225	0.700 ± 0.189	0.633 ± 0.105
migrationRule4_DS	1.0	3.0	2.0	2.0	0.000 ± 0.000	0.710 ± 0.177	0.743 ± 0.129	0.663 ± 0.151
migrationRule4_MS	1.0	4.0	2.0	3.0	0.000 ± 0.000	0.852 ± 0.077	0.652 ± 0.162	0.755 ± 0.130
migrationRule5_ALL	1.0	3.0	2.0	1.0	0.074 ± 0.163	0.709 ± 0.189	0.533 ± 0.168	0.506 ± 0.232
migrationRule5_CO	1.0	3.0	2.0	2.0	0.000 ± 0.000	0.900 ± 0.225	0.833 ± 0.236	0.933 ± 0.141
migrationRule5_DS	1.0	2.0	1.0	1.0	0.000 ± 0.000	0.913 ± 0.031	0.903 ± 0.000	0.903 ± 0.000
migrationRule5_MS	1.0	2.0	1.0	1.0	0.000 ± 0.000	0.500 ± 0.527	0.500 ± 0.527	0.600 ± 0.516
migrationRule7_ALL	1.0	3.0	4.0	2.0	0.113 ± 0.063	0.892 ± 0.049	0.920 ± 0.054	0.839 ± 0.065
migrationRule7_CO	1.0	2.0	4.0	3.0	0.083 ± 0.061	0.928 ± 0.046	0.971 ± 0.036	0.944 ± 0.055
migrationRule7_DS	1.0	3.0	4.0	2.0	0.098 ± 0.043	0.901 ± 0.065	0.936 ± 0.034	0.834 ± 0.064
migrationRule7_MS	1.0	3.0	4.0	2.0	0.091 ± 0.061	0.827 ± 0.085	0.905 ± 0.090	0.759 ± 0.129
migrationRule8_ALL	1.0	3.0	2.0	1.0	0.000 ± 0.000	0.820 ± 0.126	0.482 ± 0.155	0.500 ± 0.254
migrationRule8_CO	1.0	3.0	2.0	1.0	0.000 ± 0.000	0.778 ± 0.234	0.644 ± 0.250	0.667 ± 0.234
migrationRule8_DS	1.0	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule8_MS	1.0	3.0	2.0	2.0	0.000 ± 0.000	0.700 ± 0.205	0.517 ± 0.095	0.683 ± 0.183
Average	**1.00**	**2.66**	**2.23**	**1.69**	**0.134 ± 0.034**	**0.717 ± 0.118**	**0.664 ± 0.110**	**0.662 ± 0.129**

Table 8: Detailed instance-level results for Workflow Scheduling, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better).

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
CyberShake_100	1.0	3.0	4.0	2.0	0.076 ± 0.045	0.639 ± 0.174	0.723 ± 0.159	0.633 ± 0.143
CyberShake_1000	1.0	2.0	3.0	1.0	0.119 ± 0.090	0.755 ± 0.083	0.824 ± 0.130	0.735 ± 0.089
CyberShake_30	1.0	3.0	4.0	2.0	0.074 ± 0.092	0.284 ± 0.276	0.343 ± 0.270	0.174 ± 0.115
CyberShake_50	1.0	2.0	3.0	1.0	0.097 ± 0.086	0.684 ± 0.113	0.754 ± 0.174	0.684 ± 0.140
Epigenomics_100	1.0	3.0	4.0	2.0	0.148 ± 0.124	0.750 ± 0.120	0.823 ± 0.092	0.685 ± 0.167
Epigenomics_24	1.0	4.0	3.0	2.0	0.239 ± 0.132	0.717 ± 0.169	0.657 ± 0.110	0.622 ± 0.185
Epigenomics_46	1.0	3.0	2.0	2.0	0.052 ± 0.074	0.844 ± 0.035	0.813 ± 0.152	0.742 ± 0.191
Epigenomics_997	1.0	2.0	1.0	1.0	0.264 ± 0.177	0.769 ± 0.063	0.746 ± 0.141	0.736 ± 0.178
Inspiral_100	1.0	2.0	1.0	1.0	0.085 ± 0.059	0.718 ± 0.080	0.747 ± 0.149	0.675 ± 0.193
Inspiral_1000	1.0	2.0	3.0	3.0	0.151 ± 0.112	0.792 ± 0.079	0.835 ± 0.096	0.829 ± 0.128
Inspiral_30	1.0	2.0	3.0	1.0	0.000 ± 0.000	0.346 ± 0.331	0.390 ± 0.319	0.474 ± 0.373
Inspiral_50	1.0	2.0	3.0	1.0	0.061 ± 0.060	0.772 ± 0.144	0.796 ± 0.126	0.760 ± 0.081
Montage_100	1.0	2.0	3.0	1.0	0.064 ± 0.053	0.629 ± 0.141	0.814 ± 0.124	0.618 ± 0.203
Montage_1000	1.0	3.0	2.0	1.0	0.129 ± 0.113	0.916 ± 0.050	0.887 ± 0.040	0.888 ± 0.060
Montage_25	1.0	2.0	3.0	4.0	0.033 ± 0.015	0.816 ± 0.045	0.835 ± 0.035	0.844 ± 0.064
Montage_50	1.0	2.0	1.0	1.0	0.055 ± 0.059	0.661 ± 0.117	0.657 ± 0.184	0.638 ± 0.164
Sipht_100	1.0	2.0	3.0	3.0	0.043 ± 0.053	0.341 ± 0.283	0.531 ± 0.386	0.452 ± 0.271
Sipht_1000	1.0	2.0	4.0	3.0	0.044 ± 0.054	0.364 ± 0.172	0.668 ± 0.250	0.495 ± 0.135
Sipht_30	1.0	3.0	4.0	2.0	0.049 ± 0.064	0.745 ± 0.127	0.758 ± 0.194	0.700 ± 0.162
Sipht_60	1.0	2.0	3.0	3.0	0.074 ± 0.062	0.461 ± 0.065	0.514 ± 0.226	0.444 ± 0.103
Average	**1.00**	**2.40**	**2.85**	**1.85**	**0.093 ± 0.076**	**0.650 ± 0.133**	**0.706 ± 0.168**	**0.641 ± 0.157**

Table 9: Detailed instance-level results for Web Service Composition, showing Scott-Knott ESD ranks and the mean ± standard deviation of optimized and normalized performance (smaller is better). Blue cells denote the best Scott-Knott ESD ranks, Yellow cells denote the algorithm (among the four) with the smallest mean in that dataset.

Instance	**Rank**				**Mean ± Std**			
	MMO$_b$	**SWAY**	**LINE**	**LITE**	**MMO$_b$**	**SWAY**	**LINE**	**LITE**
workflow_1	1.0	3.0	2.0	1.0	0.132 ± 0.095	0.741 ± 0.149	0.729 ± 0.195	0.812 ± 0.111
workflow_10	1.0	3.0	2.0	2.0	0.242 ± 0.143	0.764 ± 0.168	0.710 ± 0.156	0.649 ± 0.215
workflow_10_reverse	2.0	2.0	1.0	1.0	0.671 ± 0.278	0.698 ± 0.186	0.623 ± 0.242	0.653 ± 0.161
workflow_1_reverse	1.0	3.0	2.0	1.0	0.592 ± 0.247	0.757 ± 0.162	0.754 ± 0.041	0.868 ± 0.102
workflow_2	1.0	3.0	2.0	4.0	0.196 ± 0.131	0.714 ± 0.117	0.649 ± 0.188	0.761 ± 0.175
workflow_2_reverse	1.0	3.0	2.0	2.0	0.538 ± 0.238	0.712 ± 0.178	0.707 ± 0.092	0.614 ± 0.197
workflow_3	1.0	2.0	1.0	1.0	0.257 ± 0.187	0.794 ± 0.121	0.783 ± 0.107	0.791 ± 0.107
workflow_3_reverse	1.0	3.0	2.0	1.0	0.525 ± 0.227	0.804 ± 0.126	0.750 ± 0.173	0.794 ± 0.161
workflow_4	1.0	3.0	2.0	2.0	0.111 ± 0.090	0.709 ± 0.131	0.578 ± 0.176	0.660 ± 0.210
workflow_4_reverse	1.0	2.0	1.0	1.0	0.397 ± 0.227	0.690 ± 0.155	0.705 ± 0.108	0.687 ± 0.186
workflow_5	1.0	3.0	2.0	1.0	0.212 ± 0.153	0.726 ± 0.191	0.703 ± 0.168	0.700 ± 0.239
workflow_5_reverse	1.0	2.0	3.0	3.0	0.458 ± 0.288	0.716 ± 0.104	0.757 ± 0.079	0.700 ± 0.275
workflow_6	1.0	3.0	2.0	1.0	0.133 ± 0.106	0.662 ± 0.143	0.624 ± 0.180	0.594 ± 0.151
workflow_6_reverse	1.0	3.0	2.0	2.0	0.516 ± 0.246	0.737 ± 0.133	0.774 ± 0.146	0.698 ± 0.133
workflow_7	1.0	3.0	2.0	2.0	0.188 ± 0.135	0.662 ± 0.215	0.585 ± 0.076	0.676 ± 0.159
workflow_7_reverse	1.0	2.0	3.0	3.0	0.447 ± 0.263	0.777 ± 0.193	0.845 ± 0.115	0.814 ± 0.163
workflow_8	1.0	2.0	3.0	1.0	0.274 ± 0.159	0.640 ± 0.158	0.663 ± 0.174	0.707 ± 0.162
workflow_8_reverse	1.0	1.0	2.0	2.0	0.507 ± 0.349	0.652 ± 0.230	0.611 ± 0.206	0.655 ± 0.197
workflow_9	1.0	4.0	2.0	3.0	0.205 ± 0.159	0.754 ± 0.182	0.584 ± 0.123	0.686 ± 0.128
workflow_9_reverse	1.0	3.0	2.0	2.0	0.483 ± 0.277	0.782 ± 0.146	0.783 ± 0.103	0.665 ± 0.121
Average	**1.05**	**2.65**	**2.00**	**1.80**	**0.354 ± 0.200**	**0.725 ± 0.159**	**0.696 ± 0.142**	**0.709 ± 0.168**